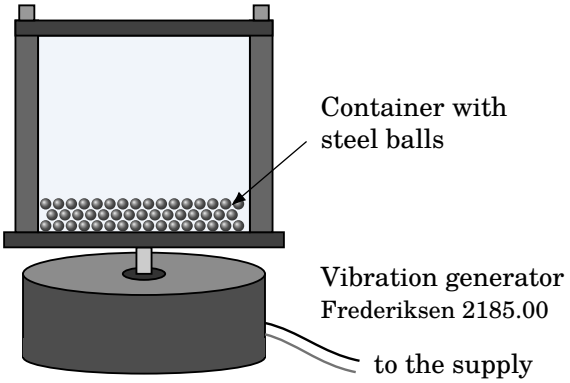
Name: _____

Class: _____

Date: _____ **Mr. Rawson • GCSE Physics**

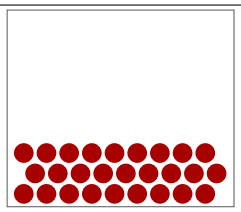
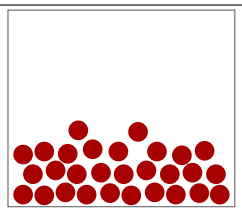
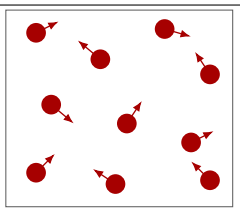
Molecular Motion Model — Demo



How the model works

- The **51 steel balls** are the **particles**.
- The vibration generator shakes the container. That **gives the balls energy**, just as **heating** gives particles energy.
- A **higher voltage** shakes harder: it models a **higher temperature**.

At each setting, draw the balls, describe how they move, and name the state.

Supply	**1 V**	**2 V**	**5 V**
Draw the balls			
How they move	*vibrate in place; the rows stay together*	*move around each other, still touching*	*fly fast in all directions, far apart*
State it models	*solid*	*liquid*	*gas*

1) **Describe** how the movement of the balls changes as the supply goes from 1 V to 5 V. **[2]**

At 1 V they only vibrate about fixed places, in rows. At 2 V they move around each other but stay touching. At 5 V they move fast in all directions and spread out to fill the container.

2) **Explain** what turning up the voltage models in a real substance. **[2]**

Heating: more energy is transferred to the particles, so they move faster (more kinetic energy). A higher voltage models a higher temperature.

3) **Explain** why the same balls take up much more space at 5 V, and what this tells you about the density of a gas. **[2]**

The balls move fast and spread far apart, so the same number of balls (the same mass) fills a much bigger volume. A gas therefore has a much lower density than a solid or a liquid.